Strategic Choices Based on Precision Service During the Construction of Unmanned Pharmaceutical Micro-Warehouses: A Dynamic Evolutionary Game Approach

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***Abstract*—Unmanned pharmaceutical micro-warehouses (UPMWs) automate operations such as warehousing, sorting, and selling by integrating the Internet of Things, artificial intelligence, and other technologies. In pharmaceutical retail settings, UPMWs help to solve problems such as purchasing medication at night, enhancing consumers' purchasing experience and reducing pharmaceutical enterprises' costs. However, the construction feasibility of UPMWs is not well researched. To bridge this gap, this article considers the precise service capabilities of UPMWs. It builds a tripartite evolutionary game model involving pharmaceutical enterprises, consumers and suppliers. It also conducts extensive robustness check using the stability test of equilibrium points, numerical simulation, and scenario expansion verification. Our major findings are as follows. First, managers of pharmaceutical enterprises exhibit a nonprofit-oriented nature, and the preference of pharmaceutical enterprise managers plays a decisive role. Second, consumers opt to use UPMWs only when the difference in utility between an UPMW and the traditional model exceeds the difference between the attention cost of consumers and added utility. Suppliers' choices depend on the disparity between upgrade costs and benefits, and their acceptable cost threshold increases as both the pharmaceutical enterprise's construction willingness and consumers' usage intention increase. Third, the precise service ability, sales proportion and acceptance of UPMWs influence the evolution rate and direction of the modelled game. Specifically, when these values are excessively low, the system evolves from 'building UPMWs' to "not building UPMWs." Overall, our findings provide a new analytical framework and practical insights for resource allocation and collaborative management of UPMW Construction.**

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***Managerial Relevance Statement*—This study provides clear practical guidance and implementation basis for the engineering management of unmanned pharmaceutical micro-warehouse (UPMW). Prior to the UPMW construction phase, managers need to balance profit objectives with decision-making preferences, establish standardized consumer demand feedback mechanisms and supplier cooperation evaluation systems, and eliminate potential risks arising from blind investment or conservative decisions. During the UPMW construction phase, managers should systematically expand the range of pharmaceutical products, optimize key nodes in the medication collection process, and effectively strengthen the core utility differences from traditional models. In the UPMW operation phase, managers need to rely on seed users such as young groups with high acceptance to build standardized word-of-mouth communication paths and fully stimulate market usage willingness. A dynamic tracking mechanism for core indicators should be established to monitor data such as precise service capabilities and sales proportions in real time; when indicators fail to meet expectations, timely initiate UPMW layout iteration, AI medication recommendation algorithm optimization, and inventory management system upgrades. Simultaneously, implement a standardized subsidy mechanism to eliminate suppliers' concerns about technological upgrading and accelerate the virtuous cycle of "enterprise construction - consumer use - supplier upgrading". This article also contributes to the following SDGs: SDG 3, SDG 9.**

***Index Terms*—Dynamic evolution, evolutionary game, strategy choice, unmanned pharmaceutical micro-warehouses (UPMWs).**

I. Introduction

An unmanned pharmaceutical micro-warehouse (UPMW) is a miniaturized intelligent pharmaceutical retail solution with warehousing functions, adopting technologies such as automation, the Internet of Things (IoT) and artificial intelligence (AI). Its goal is to achieve unmanned operation from storage and sorting to distribution and management [1], [2]. An UPMW can improve the efficiency of pharmaceutical circulation, ensure drug quality and safety, reduce reliance on manual labor and increase drug accessibility in diversified scenarios. In China, Hua Shi Pharmacy and Mei Tuan Purchasing Medicine jointly constructed an UPMW using an "Internet-enabled self-service medicine cabinet" model in May 2023. This system, which provides remote pharmacist services, operates 24/7 to respond to consumers' needs, thus overcoming the limitations of traditional self-service medicine cabinets that sell only over-the-counter drugs. Customers of traditional pharmacy services often face difficulties in obtaining prescription drugs at night due to pharmacy closures. In such cases, consumers must visit hospitals. Compared with traditional pharmacies, an UPMW can provide 24/7 services and include an AI-based medication purchasing function that recommends the appropriate drugs based on consumers' symptoms. When consumers need a doctor's assistance, an UPMW can provide a remote consultation. The medication purchasing service provided by an UPMW is more precise than that of a traditional pharmacy. In this article, we refer to these functions of an UPMW as precise services. This article focuses on the application of UPMW in the pharmaceutical field. However, the highly integrated hardware and software systems of unmanned micro-warehouses, which are built on automation, IoT and AI, possess strong scenario adaptability. Their application scope is not limited to pharmaceutical logistics. It can also be extended to fields such as food retail and beauty and personal care [3], [4].

UPMWs enable automated warehousing, sorting and inventory management through the integration of IoT and AI technologies. This highly automated operation mode significantly reduces manual operation times and error rates and thus enhances enterprises' service capabilities [5]. For example, during the pilot phase of an UPMW at the MAI Shop in Mei Tuan, the average time from order placement to fulfilment was only 3.43 min, demonstrating the considerable potential to enhance logistical efficiency. Enterprises that implement UPMWs create a win–win situation for themselves and consumers. For enterprises, UPMWs can simplify traditional pharmacy purchasing processes, reduce staffing requirements and operating costs and expand service coverage, leading to higher returns. For consumers, the precise services provided by UPMWs help to address the difficulties of purchasing medication at night and obtaining prescription drugs. The service innovations provided by UPMWs enhance the consumer shopping experience [6], [7]. Given these advantages, the implementation of UPMWs in the pharmaceutical industry has become a prominent research topic.

The precision services provided by UPMWs may enhance pharmaceutical enterprises' profitability. For example, an UPMW can store the same quantity of pharmaceuticals within 10 square meters as a traditional pharmacy can store in 100 square meters. However, as UPMWs represent a new business model, their construction remains in the exploratory stage, and the operational model and industry standards remain under development. In addition, the profitability of UPMWs remains uncertain. Therefore, this article focuses on three major engineering management issues: uncertainties in project decision, dilemmas in supply chain collaboration, and risks in user demand matching [8], [9], [10], [11]. Regarding the uncertainty in project decision: First, the benefits of UPMWs for pharmaceutical enterprises are uncertain [12], and significant investment is needed to acquire the necessary technology and equipment. The primary goal of UPMW construction is to improve the service capacity and thus generate higher returns. It remains unclear whether pharmaceutical enterprises are sufficiently motivated to invest in this construction. Therefore, it is crucial to determine when pharmaceutical enterprises are willing to build UPMWs. Second, pharmaceutical enterprises that build UPMWs often also continue to operate their traditional retail sales model. Determining the optimal allocation between traditional sales and UPMW-based distribution remains a key challenge. Third, UPMW operations rely on unmanned equipment and associated technology. To remain competitive and attract consumers, pharmaceutical enterprises must continuously improve UPMWs' precision service capabilities. This optimization requires suppliers to provide ongoing technological advancements and equipment upgrades. However, it remains unclear how to coordinate the interests of pharmaceutical enterprises with those of suppliers to ensure that the latter have sufficient incentives to invest in technological upgrades.

Regarding the risks in consumers demand matching, encouraging consumers to accept UPMWs can be challenging. In the early stage of UPMW implementation, the limited precision service capabilities may discourage adoption. Pharmaceutical enterprises cannot accurately predict consumer behavior, which introduces the risk of revenue losses. Furthermore, if a pharmaceutical enterprise chooses not to build UPMWs when its competitor has done so, it may suffer from decreases in both service capacity and market share. Therefore, it is crucial to identify when consumers are willing to accept and use UPMWs.

Finally, regarding the dilemmas in supply chain collaboration, suppliers sell the equipment and technology required for UPMWs to pharmaceutical enterprises, and thus suppliers' profits depend on pharmaceutical enterprises' acceptance. First, suppliers must be committed to enhancing the enterprises' service capabilities by upgrading UPMW equipment and related technologies; however, R&D investment is costly [13]. Hence, suppliers must weigh the benefits and costs of equipment and technological upgrades. Second, the requirements for suppliers' technology upgrading call for the close collaboration of pharmaceutical enterprises. Even in the early stage of suppliers' technology upgrading, pharmaceutical enterprises need to provide necessary financial support to achieve technology upgrading through collaborative innovation [14].

During the construction of an UPMW, pharmaceutical enterprises and suppliers should focus on enhancing their precision service capabilities to attract more consumers. Therefore, it is crucial to examine the strategic choices of suppliers, pharmaceutical enterprises and consumers during UPMW construction.

While unmanned retail has been widely adopted worldwide [15], research on UPMWs is rather limited. Most studies in this area focus on unmanned retail cabinets and unmanned supermarkets [16], [17], [18], [19], as well as on technological adoption and optimization. The further development of unmanned technology has resulted in fully unmanned retail becoming trendy. As an emerging business model, UPMWs differ substantially from unmanned vending cabinets, with integrated features such as automated replenishment, delivery and remote consultations. However, the limited research on UPMWs largely focuses on unmanned stores, with little attention given to the feasibility of UPMWs and stakeholders' strategic choices.

UPMW construction is a dynamic process. Participants' strategic choices affect this process and change over time. Accordingly, we pose the following questions in this article: How do the decision-making processes of pharmaceutical enterprises, consumers and suppliers evolve during UPMW construction?

How do their strategic choices affect the overall system? Which party holds a dominant role in the tripartite game process? How do key parameters influence this game process?

In summary, the UPMW is an emerging business format intended to improve the efficiency of drug purchasing while reducing labor costs [20]. Construction entails complex interactions among pharmaceutical enterprises, suppliers, and consumers. Notably, the strategic choices of each participant with respect to technology adoption and equipment upgrading exhibit distinct characteristics of dynamic evolution. System dynamics and game theoretical models such as the Stackelberg game, which assume perfect rationality and static equilibrium, struggle to characterize the strategic adjustment behaviors of the participants. These behaviors arise from information asymmetry, bounded rationality and environmental uncertainties. Accordingly, in this article, we apply an evolutionary game to include bounded rationality and group dynamic interaction mechanisms and explore the dynamic evolution of UPMW construction and corresponding strategic choices from the perspectives of pharmaceutical enterprises, consumers and suppliers.

Through the aforementioned research, this study has addressed the research questions proposed herein and obtained the following conclusions. First, when managers are aggressive, pharmaceutical enterprises firmly support UPMWs. When managers are conservative, pharmaceutical enterprises tend not to establish UPMWs, regardless of their profitability. Second, consumers use an UPMW only when the utility difference between the UPMW and the traditional model exceeds the gap between the attention cost and potential additional utility. Suppliers' decisions are determined by weighing the costs and benefits, and their acceptable cost threshold increases as both pharmaceutical enterprises' construction willingness and consumers' usage intention also increase. Third, the precise service ability, sales proportion and acceptance of UPMWs influence the evolution rate and direction of the game system. Specifically, when these values are excessively low, the system evolves from "building UPMWs" to "not building UPMWs."

The remainder of this article is organized as follows. Section II presents a literature review. Section III describes the problem and parameter settings. Section IV analyzes evolutionary equilibrium strategies. Section V focuses on simulation analysis. Section VI concludes the research and discusses potential managerial insights.

II. Literature Review

A. Unmanned Retail and UPMW

With the advancement of technology and the development of the retail industry, "unmanned retail" has been widely implemented on a global scale [4], [21], [22]. Using advanced technologies such as the IoT and big data [23], unmanned retail enables self-service purchasing, payment and inventory monitoring, thus reducing the retail industry's dependence on manual labor. Unmanned retail, also known as unmanned stores or self-service stores, is characterized by automated operations, customer self-service, and a data- and technology- driven model in which daily operations are performed through self-checkout systems, smart shelves and cameras [18]. Unmanned retail has diversified to include unmanned vending machines, convenience stores and many other applications. For instance, unmanned vending machines can be found in parks, stations, and stadiums, and there are also cases like JD X Unmanned Supermarkets in China.

Important research on the operation and related technical challenges of unmanned retail has been conducted. Some scholars have examined the use of technologies such as the IoT in unmanned retail and conducted in-depth research on product identification methods and the optimization of unmanned retail. For example, Jeon et al. [24] designed a vision-based unmanned retail system that could identify products accurately. Wang et al. [25] introduced a large language model to enhance the service capability of unmanned retail stores by addressing the stores' inability to provide shopping suggestions. Other scholars have focused on practical aspects of unmanned retail, such as consumers' acceptance and preferences and the replenishment cycle. Using a sample of 600 participants, Nam et al. [18] examined consumers' preferences for store- and technology-related attributes in the context of retail goods and services. Zheng and Li [26] predicted purchase demand based on an analysis of historical sales data and implemented supply chain optimization for unmanned retail stores.

Unmanned retail has gradually become a major direction of development in the retail industry, and many scholars have validated the advantages of unmanned retail over traditional retail mode from different perspectives. However, these studies often focus on unmanned stores and vending machines, while limited research has explored UPMWs in the pharmaceutical industry. The concept of an UPMW first appeared in 2019, when Mei Tuan Flash Purchase in China introduced an unmanned micro-warehouse where retail home scene orders could be automatically completed through micro-pre-warehouses. UPMWs automate the entire process, including product recommendation, online ordering, intelligent shelf picking, AGV robot transportation, automatic verification, packaging and distribution, thus creating integrated closed-loop merchant services. However, research has not adequately addressed the strategic choice used by pharmaceutical enterprises, suppliers and consumers in the construction and operation of UPMWs.

B. Evolutionary Game Theory and Enterprise Decision

Evolutionary game theory (EGT) combines game theory with dynamic evolutionary process analysis and can be used to study both competitive and cooperative behaviors and strategic choices. This research method is commonly used to explore the impact of enterprise behavior on stakeholders, including technological innovation [27], market competition [28], [29], [30], cooperation and alliance [31], [32], [33], and green innovation [34], [35]. For example, Gong and Dai [36] constructed an evolutionary game model combining government, enterprises and consumers and explored decision-making and choices related to enterprises' green technological innovation. Liu et al. [37] applied EGT to a two-level green supply chain that consisted of green suppliers and green manufacturers. The EGT model examines the various internal and external factors affecting the behavior of both parties in the game and analyses the evolution and stability of collaborative emission reduction using numerical simulations. Therefore, EGT offers significant advantages in terms of addressing strategic choice problems, as it can identify the best strategy.

UPMW construction and operation involve three main participants: pharmaceutical enterprises, suppliers and consumers. System stability is achieved through multiple iterations based on the participants' strategic choices. We applied an evolutionary game in the research for three reasons. First, system dynamics and classical game theoretical models like the Stackelberg model assume that participants are completely rational and information is symmetrical, and that they pursue the maximization of their own interests [38]. However, in reality, participants (such as pharmaceutical enterprises or suppliers) have different perceptions of technology's costs and risks. Participants' decisions must be gradually adjusted through trial-and-error and imitation and cannot be completely rational. EGT, which is based on the idea of biological evolution and bounded rationality [39], posits that participants' rationality and ability are flawed, a more realistic assumption than that of other game theoretical models. Second, according to EGT, a group of participants are the research object, and the focus is the process by which group behavior reaches equilibrium, rather than the optimal decision of a single participant. However, game theoretical models often explore a single entity with clear boundaries and independent goals. Therefore, EGT is more in line with the reality of the interactions between multiple participating groups and their and co-evolution during UPMW construction. Third, UPMWs construction is a long-term interactive endeavor in which technology promotion requires a prolonged dynamic adjustment rather than a one-time game. EGT emphasizes the dynamic nature of economic change and the system's tendency to move from one equilibrium state to another. As such, an EGT model can be more effective than other game theoretical models [40]. Methods such as system dynamics cannot reflect the transition process of the system from one equilibrium state to another, nor can they fully reflect the impact of one party's decisions on the other party's decisions. In summary, we choose to explore the dynamic evolution of UPMW construction and operation from the perspective of EGT, while considering pharmaceutical enterprises, suppliers, and consumers.

C. Literature Comparison

Unmanned retail is the focus of extensive research, and its role has gained wide academic recognition. The use of EGT to model decision-making dynamics is prominent in corporate contexts. However, some gaps in UPMW research remain. First, the literature is predominantly centered on unmanned stores and vending cabinets. Studies related to pharmaceuticals often have emphasized pricing strategies and drug quality control. Few studies have addressed the applications of UPMW in pharmaceutical retail settings, and relevant studies on the feasibility of UPMWs are lacking. Second, most research related to unmanned retail has tended to explore the impact of unmanned retail stores on consumers after construction is completed, how to improve the consumer experience and the factors that affect consumer use. Problems related to constructing an UPMW have not been discussed. Given that UPMWs are a new initiative in the pharmaceutical retail industry, the factors affecting their construction by pharmaceutical enterprises remain unclear. Third, two critical questions about this emerging business model remain unexplored: 1) How do consumers' willingness to adopt and equipment suppliers' willingness to upgrade evolve? 2) How can these respective forms of willingness be enhanced? To address these research gaps, we use a tripartite evolutionary game approach to analyze the strategic choices made during UPMW construction. A comparative analysis of the relevant literature is summarized in Table I.

III. Model Background and Parameter Setting

A. Model Background

From the perspective of stakeholder collaboration in engineering management, the construction of UPMWs is an engineering project involving multiple entities. Its core goal is to achieve the dual objectives of maximizing benefits and user value through technology integration and process optimization. As the initiators of the project, pharmaceutical enterprises need to comprehensively evaluate the costs and benefits of project construction, while considering consumers' acceptance of precision services and suppliers' technical support capabilities. From a theoretical perspective, this emerging business model offers certain advantages over traditional pharmacies [7]. During actual operation, a pharmaceutical enterprise must fully consider the interactions among stakeholders to make optimal decisions. By drawing on EGT, we can account for the interests of multiple stakeholders and address conflicts and cooperation among them. We use a tripartite EGT framework to explore the interactions among pharmaceutical enterprises, suppliers and consumers and focus on decision-making that enables the formation of a stable system. Pharmaceutical enterprises are the dominant players in UPMW construction, while suppliers and consumers are the key stakeholders.

In an increasingly competitive world, enhancing service quality has emerged as a crucial strategy for elevating customer satisfaction and encouraging loyalty. The aim of precision service is to maximize the consumer's experience by addressing their core problems, thereby increasing their loyalty and satisfaction [41]. At present, pharmaceutical retail consumers face challenges such as difficulties in purchasing medication at night and the inability to self-purchase prescription drugs. Pharmaceutical enterprises cannot address these issues simply by opening more traditional pharmacies, as this would increase their operating costs. Therefore, the construction of UPMWs based on precision service can help enterprises to address these issues. For example, Fahrenheit Pharmacy's UPMW can offer over 1000 different medications in a 10-square-meter space. UPMWs do not require manual sorting and can provide remote consultations, thus addressing the problem of purchasing prescription drugs. Such services can improve consumers' satisfaction and help businesses to increase their profits.

TABLE I
COMPARISON OF THE CURRENT STUDY WITH THE MOST RELEVANT LITERATURE

Contents	Research issues	Research methods	Research subject	Research of corporate measures	Scope of research
Mostofi et al., 2024	A new pricing mechanism for pharmaceutical supply chains: a game theory analytical approach for healthcare service	Game theory	Government, enterprises	Drug pricing	Non specific country
Zhang and Zhu, 2021	Coregulation supervision strategy of drug enterprises under the government reward and punishment mechanism	Evolutionary game	Drug enterprises, government, third-party testing institutions	Coregulatio n supervision strategy	Non specific country
Nouri-Harzvili et al., 2024	Evolutionary marketing strategies for new high-technology product sales: Effects of customers' innovation adoption	Evolutionary game	Distributors, clients	New high-technology product Sales	Non specific country
Nam et al., 2024	Consumer preferences for unmanned stores: A choice experiment study	Empirical analysis	Consumer	Consumer preferences	Korea
Zheng et al., 2018	Unmanned retail's distribution strategy based on sales forecasting	Optimization	Unmanned retail	Distributio n strategy	Non specific country
This study	Strategic choices during construction of UPMWs based on precision service: a dynamic evolutionary game approach	Evolutionary game	Pharmaceutic al enterprises, suppliers, consumers	Feasibility of UPMW constructio n	Non-specific country

UPMWs are intended to address the inherent limitations of the traditional pharmaceutical retail industry, enhance the consumer experience and improve customer satisfaction, leading to higher revenues. An UPMW's revenue is based on precision service and depends heavily on customer acceptance; therefore, we assume that it is affected by customer acceptance. We introduce customer acceptance into the model as an index to measure the precision service capabilities of an UPMW. When customer acceptance decreases below a threshold, consumers will refuse to use UPMWs, which will negatively affect the interests of enterprises.

B. Parameter Settings

To explore the impact of the strategic choices of pharmaceutical enterprises, consumers and suppliers on the construction and subsequent operation of UPMW systems, we construct a tripartite evolutionary game model. Consumers, pharmaceutical

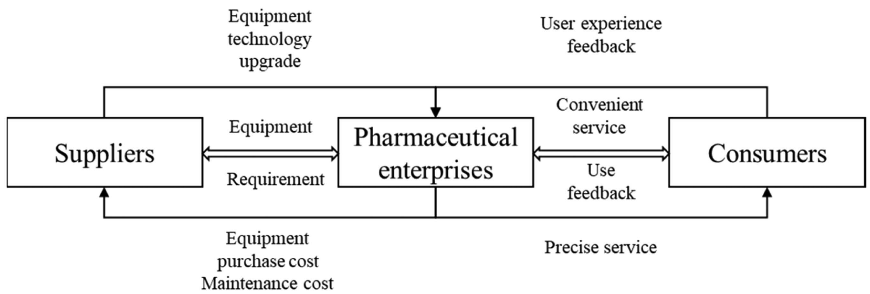


Fig. 1. Tripartite evolutionary game structure representing an UPMW.

enterprises and suppliers are the central players in UPMW construction and use. In this interaction, suppliers, pharmaceutical enterprises and consumers each seek to maximize the benefits of their decisions based on the resources at their disposal. Therefore, when we discuss strategic choices related to UPMW construction and application, suppliers, pharmaceutical enterprises, and consumers are the main players selected for in-depth analysis, as shown in Fig. 1.

Specifically, suppliers are the group responsible for building UPMW equipment and can offer two strategies to maximize their own benefits: upgrading or not upgrading the UPMW facilities and equipment. Pharmaceutical enterprises are entities with the ability to construct UPMWs and provide related services. However, these enterprises are driven by the pursuit of profit maximization and must carefully assess market demand, then decide whether to invest resources in UPMW construction based on actual consumer demand for UPMW services. This strategic choice process fully reflects the flexibility of pharmaceutical enterprises when faced with market changes. Consumers are those who must decide whether to adopt UPMWs or continue with traditional shopping models. When choosing, consumers consider factors such as convenience, price advantages and personal shopping habits. Their decisions not only affect UPMW acceptance and market penetration but also indirectly influence enterprises' construction strategies.

Assumption 1: The participants have a limited prediction ability. Completely rational decision-making is difficult due to incomplete information, various constraints, cognitive biases and other factors. Therefore, the three parties' strategic choices regarding UPMW adoption should exhibit bounded rationality [42]. In addition, information concerning the decisions of other players is asymmetric in the initial stage of the game [43], [44].

Assumption 2: UPWM construction and use requires the joint participation of suppliers, pharmaceutical enterprises and consumers. Participants constantly adjust their own strategies according to the strategic choices of the other participants until stability is achieved. In this three-party game, suppose that the probability of a pharmaceutical enterprise choosing to build an UPMW is x $(0 \leq x \leq 1)$. The probability of the pharmaceutical enterprise choosing not to build an UPMW is therefore $1 - x$. The probability of consumers choosing to adopt the UPMW strategy is y $(0 \leq y \leq 1)$. The probability of consumers choosing a traditional shopping model strategy is $1 - y$. The supplier can choose between two strategies: to upgrade or not to upgrade equipment. The probability of suppliers choosing to upgrade is z $(0 \leq z \leq 1)$, and the probability of choosing not to upgrade is $1 - z$.

Assumption 3: In the game, cost and benefit are set in accordance with prior literature [45], [46], [47]; we incorporate the UPMW settings of Fahrenheit Pharmacy in China with the following assumptions. Fahrenheit Pharmacy operates both traditional pharmacy services and UPMW services simultaneously and thus needs to balance the sales of the two channels. Assume that the proportion of products sold through UPMWs is T $(0 \leq T \leq 1)$. If the pharmaceutical enterprise and the consumer choose to adopt the UPMW at the same time, the pharmaceutical enterprise obtains the benefits TP_2 from the UPMW and $(1 - T)P_1$ from the traditional channel. Pharmaceutical enterprises must pay the cost of the input needed to build the UPMW and the cost of operating the traditional business C_2. If the pharmaceutical enterprise chooses to construct an UPMW and the consumer chooses not to use it, the pharmaceutical enterprise only gains a revenue of $(1 - T)P_1$. Even though TP_2 cannot be earned, the enterprises still must pay the cost of constructing the UPMW. If the pharmaceutical enterprise chooses not to build an UPMW, it can only benefit from the traditional channel P_1. However, the enterprise must pay an additional opportunity cost L and C_2, L denotes the potential loss incurred by pharmaceutical enterprises due to missing out on the UPMW market. If the UPMW is built, its precise service capabilities may not be universally accepted, and consumers' acceptance of this new business model will also affect the UPMW's revenue. Therefore, precise service acceptance, *K*, is introduced as a measure of consumers' willingness to try and accept UPMWs. Thus, the benefits of enterprises can be represented by KTP_2. This phenomenon is reflected in the MAI shop in Mei Tuan, where some consumers are accustomed to traditional shopping or choose traditional models because of their concerns about new technologies.

Assumption 4: During the game, if both the consumers and pharmaceutical enterprise choose to adopt an UPMW, consumers will gain the utility of using the UPMW, V_2, and will pay the cost Q_2 required for the UPMW's services. If the consumer chooses to adopt an UPMW but the enterprise chooses not to build it, the consumer will not have access to UPMW services at this time. The consumer will only receive utility V_1 from the traditional model and will pay the cost of the traditional model, Q_1, and the attention cost to UPMW, B, B denotes the attention cost invested by consumers in UPMW and the utility loss resulting from their inability to use UPMW. If the consumer chooses not to adopt the UPMW strategy, they will continue to use the traditional model, resulting in a total utility of $V_1 - Q_1$.

Assumption 5: During UPMW construction, Fahrenheit Pharmacy in China must purchase UPMW equipment from its suppliers. When the UPMW is operating, malfunctions may occur, and the pharmaceutical enterprise must pay equipment maintenance fees to the suppliers. Therefore, the supplier obtains revenue M_1 from equipment sales and the equipment maintenance fee R and bears the cost C_2 required for the UPMW equipment. If the pharmaceutical enterprise chooses the non-construction strategy, suppliers do not obtain profits and must bear the UPMW equipment cost C_2.

Assumption 6: If the supplier chooses to upgrade its equipment, the UPMW's service capabilities improve, and its precise service capability ρ also increases. Researchers focused on service capacity evaluation have distinguished internal capacity evaluation [48], [49] from external customer reviews [50], [51]. In an UPMW, precision service is intended to improve the service efficiency and shopping experience; therefore, ρ can be measured based on the internal service efficiency and external service evaluation. We thus define the precise service capability as $\rho = \frac{\alpha E_1}{E_2} + \frac{\beta \mathrm{SE}_1}{\mathrm{SE}_2}$, where E_1 and E_2 represent the shopping efficiencies of the UPMW after and before the equipment upgrade, respectively, and SE_1 and SE_2 represent the consumer's shopping experience after and before the equipment upgrade, respectively. Consumers obtain $(1+\rho)V_2$ utility from using the UPMW, and suppliers pay cost C_3 for UPMW upgrades. Simultaneously, suppliers and pharmaceutical enterprises earn additional incomeΔP_2 and ΔP_1, respectively. Both ΔP_1 and ΔP_2 are related to ρ, such that $\Delta P_1 = \rho Z_1$, $\Delta P_2 = \rho Z_2$ and Z_1, Z_2 respectively represent the additional benefits gained by pharmaceutical enterprises and suppliers when the precision service capability doubles after equipment upgrading. Amazon's Just Walk Out unmanned retail store provides an example. This store integrates AI, radio frequency identification, cameras and sensors to fully automate checkouts. This approach not only enhances the consumer shopping experience and reduces labor costs but also enables suppliers to provide Amazon with related technical equipment and thus increase their revenues from large-volume orders. Table II lists the game parameters.

IV. Evolutionary Game Equilibrium Strategy Analysis

A. Model Construction

The payoffs of UPMW adoption for the pharmaceutical enterprise, consumer and supplier, based on the equilibrium solution of the game model, are summarized in the income matrix in Table III.

We construct the replication dynamic equations representing the behavioral strategies of the pharmaceutical enterprise, supplier and consumer, respectively. The expected return of a pharmaceutical enterprise choosing to build an UPMW is E_{11}, while its expected return after choosing not to build an UPMW is E_{12}, as shown in the following:

$$\begin{aligned} E_{11} =\ & yz\left[(1-T)P_1 + TKP_2 + \Delta P_1 - M_1 - C_1 - R\right] \\ & + y(1-z)\left[(1-T)P_1 + TKP_2 - M_1 - C_1 - R\right] \\ & + (1-y)z\left[(1-T)P_1 - M_1 - C_1 - R\right] \\ & + (1-y)(1-z)\left[(1-T)P_1 - M_1 - C_1 - R\right] \end{aligned} \tag{1}$$

$$E_{12} = P_1 - L - C_1. \tag{2}$$

The expected benefit of the pharmaceutical enterprise is shown as follows:

$$\overline{E_1} = xE_{11} + (1-x)E_{12}\,. \tag{3}$$

The replication dynamic equation representing the pharmaceutical enterprise's strategic behavior is as follows:

$$\begin{aligned} F(x) &= \frac{dx}{dt} = x\left(E_{11} - \overline{E_1}\right) \\ &= x(1-x)(L - M_1 - R - TP_1 + yz\Delta P_1 + yKTP_2)\,. \end{aligned} \tag{4}$$

The expected income of a consumer who chooses to use is E_{21}, and the expected income of one who chooses not to use is E_{22}, as shown in the following:

$$\begin{aligned} E_{21} &= xz\left((1+\rho)V_2 - Q_2\right) + (1-x)z(V_1 - Q_1 - B) \\ &\quad + x(1-z)(V_2 - Q_2) + (1-x)(1-z)(V_1 - Q_1 - B) \end{aligned} \tag{5}$$

$$E_{22} = V_1 - Q_1\,. \tag{6}$$

The expected benefit of the consumer is shown in the following:

$$\overline{E_2} = yE_{21} + (1-y)E_{22}. \tag{7}$$

The replication dynamic equation representing the consumer's strategic behavior is as follows:

$$\begin{aligned} F(y) &= \frac{dy}{dt} = y\left(E_{21} - \overline{E_2}\right) \\ &= y(1-y)(xB - B + xQ_1 - xQ_2 - xV_1 + xV_2 + xz\rho V_2)\,. \end{aligned} \tag{8}$$

The expected revenue of the supplier who chooses to upgrade equipment is E_{31}, and that of the supplier who chooses not to upgrade equipment is E_{32}, as shown in the following:

$$\begin{aligned} E_{31} &= xy(M_1 - C_2 - C_3 + \Delta P_2 + R) + (1-x) \\ &\quad \times y(-C_2 - C_3) + x(1-y)(M_1 - C_2 - C_3 + R) \\ &\quad + (1-x)(1-y)(-C_2 - C_3) \end{aligned} \tag{9}$$

$$\begin{aligned} E_{32} =\ & xy(M_1 - C_2 + R) + x(1-y)(M_1 - C_2 + R) \\ & + (1-x)y(-C_2) + (1-x)(1-y)(-C_2)\,. \end{aligned} \tag{10}$$

The expected benefit of the supplier is shown in the following:

$$\overline{E_3} = zE_{31} + (1-z)E_{32}. \tag{11}$$

The replication dynamic equation representing the supplier's strategic behavior is as follows:

$$F(z) = \frac{dz}{dt} = z\left(E_{31} - \overline{E_3}\right) = z(z-1)(C_3 - xy\Delta P_2)\,. \tag{12}$$

TABLE II
LIST OF GAME PARAMETERS

Parameters	Definitions
P_1	Pharmaceutical enterprises completely distribute traditional business income
P_2	Pharmaceutical enterprises fully distribute UPMW revenue
C_1	Cost of the traditional pharmaceutical enterprise business mode
K	Consumer acceptance of UPMWs
ρ	UPMW precision service capability
V_1	Utility consumers obtain from using the traditional business model
V_2	Utility consumers obtain from using UPMWs
Q_1	Cost to consumers due to using the traditional business model
Q_2	Cost to consumers of UPMWs
T	Proportion of drugs sold by a pharmaceutical enterprise in UPMWs
L	Opportunity cost paid by pharmaceutical enterprises due to not building UPMW
B	Cost of consumers to pay attention to UPMWs
M_1	UPMW suppliers' profit due to the pharmaceutical enterprise's decision to build
R	Maintenance fee that the pharmaceutical enterprise must pay
C_2	Supplier's cost for manufacturing UPMW equipment
C_3	Supplier's cost related to the decision to upgrade equipment
ΔP_1	Additional revenue of the pharmaceutical enterprise due to the supplier's upgrades
ΔP_2	Additional revenue earned by the supplier from upgrading the equipment
Z_1	Additional benefits to pharmaceutical enterprises for each doubling of precision service capabilities
Z_2	Additional benefits to suppliers from the improved unit precision service capabilities

B. Stability of the Pharmaceutical Enterprises' Strategic Choice

The impact of a pharmaceutical enterprise's probability of choosing to build an UPMW on the stability of its equilibrium strategy can be represented as follows:

$$\frac{\partial F(x)}{\partial x} = (1-2x)\left(yz\Delta P_1 + yKTP_2 + L - M_1 - R - TP_1\right). \tag{13}$$

Proposition 1: When a pharmaceutical enterprise's opportunity cost of not building an UPMW is large, i.e., $L > M_1 + R + TP_1$, the system stabilizes at $x = 1$; when this cost is small, i.e., $L < M_1 + R + TP_1 - \Delta P_1 - KTP_2$, it stabilizes at $x = 0$; when this cost is moderate, i.e., $M_1 + R + TP_1 - \Delta P_1 - KTP_2 < L < M_1 + R + TP_1$, x increases as y and z increase.

Proposition 1 indicates that the pharmaceutical enterprise's strategic inclination regarding development affects the construction of the UPMW. When the managers of the pharmaceutical enterprise are very conservative or do not pay attention to innovation, they will resolutely refuse to build an UPMW; if they are more aggressive and innovation-focused, they will firmly

TABLE III
INCOME MATRIX OF THE TRIPARTITE GAME

Supplier	Pharmaceutical enterprise	Consumer: UPMW(y) is used	Consumer: UPMW is not used $(1-y)$
Upgrade equipment (z)	UPMW construction (x)	$M_1-C_2-C_3+\Delta P_2+R$; $(1-T)P_1+TKP_2+\Delta P_1-M_1-C_1-R$; $(1+\rho)V_2-Q_2$	$M_1-C_2-C_3+R$; $(1-T)P_1-M_1-C_1-R$; V_1-Q_1
	No UPMW construction $(1-x)$	$-C_2-C_3$; P_1-L-C_1; V_1-Q_1-B	$-C_2-C_3$; P_1-C_1-L V_1-Q_1
No upgrade equipment$(1-z)$	UPMW construction (x)	M_1-C_2+R; $(1-T)P_1+TKP_2-M_1-C_1-R$; V_2-Q_2	M_1-C_2+R; $(1-T)P_1-M_1-C_1-R$; V_1-Q_1
	No UPMW construction $(1-x)$	$-C_2$; P_1-L-C_1; V_1-Q_1-B	$-C_2$; P_1-C_1-L V_1-Q_1

choose to build an UPMW even if the enterprise will face losses in the early implementation stage. This phenomenon is due to the influence of L, the enterprise's opportunity cost of not building an UPMW, which directly reflects the level of importance that the enterprise places on investing in an UPMW. The threshold of L is negatively affected by factors such as the UPMW's revenue, sales proportion and upgrade revenue and positively related to the UPMW's cost and maintenance cost and the revenue of the traditional model.

C. Stability of the Consumer's Strategic Choice

The impact of the consumer's probability of choosing to use the UPMW on the stability of their equilibrium strategy is as follows:

$$\frac{\partial F(y)}{\partial y} = (1-2y)(xB - B + xQ_1 - xQ_2 - xV_1 + xV_2 + xz\rho V2). \tag{14}$$

Proposition 2: Let $A = Q_1 - Q_2 - V_1 + V_2$, where A represents the difference between the benefits gained by consumers using UPMW and those gained when using traditional services. When $A > \frac{(1-x)B}{x} - z\rho V_2$, the strategic choice is stable at $y = 1$. When $A < -z\rho V_2$, it is stable at $y = 0$. When $-z\rho V_2 < A < \frac{(1-x)B}{x} - z\rho V_2$, y increases as $x\ and\ z$ increase.

The main factors influencing the consumer's strategic choices are the cost of their attention to the UPMW and the benefits derived from technological upgrades. For the consumer, when the sum of the utility gained from using the UPMW and the potential additional utility provided by the UPMW upgrades is less than the utility obtained from using the traditional model, using the UPMW will result in a poor experience, and consequently, the consumer will abandon the UPMW. When the utility difference between the UPMW and the traditional model exceeds the gap between the consumer's cost of attention to the UPMW and the additional utility from potential upgrades, the consumer will adopt the UPMW. When the utility difference is moderate, the threshold determining the consumer's use of the UPMW will decreases as both x and z increase. In other words, the consumer's willingness to use an UPMW increases with the pharmaceutical enterprise's propensity to construct UPMWs and the supplier's propensity for technological upgrades of the UPMW's equipment.

D. Stability of the Supplier's Strategic Choice

The impact of a supplier's probability of choosing to upgrade equipment on the stability of its equilibrium strategy can be represented as follows:

$$\frac{\partial F(z)}{\partial z} = (1-2z)(xy\Delta P_2 - C_3). \tag{15}$$

Proposition 3: When the supplier's equipment upgrading cost is small, $C_3 < xy\Delta P_2$, its strategic choice is stable at $z = 1$. When the supplier's equipment upgrading cost is large, $C_3 > xy\Delta P_2$, it is stable at $z = 0$. The threshold of equipment upgrading cost that the supplier can accept, z, increases with increasing x and y.

According to Proposition 3, the disparity between the cost and the benefit of the upgrade is the core factor influencing the supplier's choice. When the upgrade cost exceeds the upgrade benefit, the supplier opts not to upgrade the equipment. Conversely, when the upgrade cost is lower than the upgrade benefit, the

TABLE IV
EQUILIBRIUM POINTS OF THE EVOLUTIONARY GAME AND THEIR EIGENVALUES

Equalizsation point	λ_1	λ_2	λ_3
$E_1(0,0,0)$	$-1/(M_1 - L - R + TP_1)$	$-1/B$	$-1/C_3$
$E_2(1,0,0)$	$1/(M_1 - L + R + TP_1)$	$1/(Q_1 - Q_2 - V_1 + V_2)$	$-1/C_3$
$E_3(0,1,0)$	$-1/(M_1 - L + R - KTP_2)$	$1/B$	$-1/C_3$
$E_4(0,0,1)$	$-1/(M_1 - L - R + TP_1)$	$-1/B$	$1/C_3$
$E_5(1,1,0)$	$1/(M_1 - L + R + TP_1 - KTP_2)$	$-1/(Q_1 - Q_2 - V_1 + V_2)$	$-1/(C_3 - \Delta P_2)$
$E_6(1,0,1)$	$1/(M_1 + R - L + TP_1)$	$1/(Q_1 - Q_2 - V_1 + V_2 + \rho V_2)$	$1/C_3$
$E_7(0,1,1)$	$1/(L - M_1 - R - TP_1 + KTP_2)$	$1/B$	$1/C_3$
$E_8(1,1,1)$	$-1/(L - M_1 - R - TP_1 + TKP_2 + \Delta P_1)$	$-1/(Q_1 - Q_2 - V_1 + V_2 + \rho V_2)$	$1/(C_3 - \Delta P_2)$

supplier chooses to upgrade the equipment. Therefore, the three parties mutually reinforce each other through their willingness to participate. Any single participant is positively affected by the other participants, and their income is affected by the marginal benefits gained by the other participants.

E. System Asymptotic Stability Analysis

According to the equilibrium strategy, to ensure that the consumers, suppliers and pharmaceutical enterprises in the tripartite evolutionary game have a stable strategy, the following requirements must be met:

$$\begin{cases} \frac{dx}{dt} = 0 \\ \frac{dy}{dt} = 0 \\ \frac{dz}{dt} - 0 \end{cases}. \tag{16}$$

The solution shows eight special equilibrium points among the game participants: (0,0,0), (0,1,0), (0,0,1), (1,0,0), (0,1,1), (1,0,1), (1,1,0), and (1,1,1).

According to equilibrium theory, the asymptotic stability of the equilibrium point is determined by the signs on the eigenvalues of the system's Jacobian matrix. To satisfy Evolutionary Stable Strategy (ESS), the eigenvalues of the Jacobian matrix of the system must all be negative; this condition is necessary and sufficient [52]. Therefore, according to (13), the Jacobian matrix of the tripartite game system can be expressed as follows:

$$J\ (x,y,z) = \begin{pmatrix} \frac{\partial F(x)}{\partial x} & \frac{\partial F(x)}{\partial y} & \frac{\partial F(x)}{\partial z} \\ \frac{\partial F(y)}{\partial x} & \frac{\partial F(y)}{\partial y} & \frac{\partial F(y)}{\partial z} \\ \frac{\partial F(z)}{\partial x} & \frac{\partial F(z)}{\partial y} & \frac{\partial F(z)}{\partial z} \end{pmatrix}$$

$$= \begin{pmatrix} J_{11} & J_{12} & J_{13} \\ J_{21} & J_{22} & J_{23} \\ J_{31} & J_{32} & J_{33} \end{pmatrix} \tag{17}$$

$J_{11} = x(R - L + TP_1 + M_1 - yz\Delta P_1 - yTKP_2) + (x-1)(R - L - P_1T + M_1 - yz\Delta P_1 - yTKP_2)$, $J_{12} = -x(x-1)(z\Delta P_1 + KTP_2)$, $J_{13} = -x(x-1)\,y\,\Delta P_1$, $J_{21} = y(1-y)(B + Q_1 - Q_2 - V_1 + V_2 + \rho V_2)$, $J_{22} = (y-1)\ (B - xB - xQ_1 + xQ_2 + xV_1 - xV_2 - xz\rho V_2) + y(B - xB - xQ_1 + xQ_2 + xV_1 - xV_2 - xz\rho V_2)$, $J_{23} = -y(y-1)\,x\rho V_2$, $J_{31} = -z(z-1)\,y\Delta P_2$, $J_{32} = -z(z-1)\,x\Delta P_2$, $J_{33} = (z-1)(C_3 - xy\Delta P_2) + z(C_3 - xy\Delta P_2)$.

The corresponding Jacobian matrix eigenvalues can be obtained by substituting the system equilibrium point. For example, when $E_1 = (0,0,0)$, the corresponding eigenvalues are $0,\ -B,\ L - B - P_1T - NM_2$. The dynamic Jacobian matrix eigenvalues obtained via substituting eight equilibrium points in turn are shown in Table IV.

According to Table IV, the characteristic values of the four local equilibrium points $E_3(0,1,0)$, $E_4(0,0,1)$, $E_6(1,0,1)$ and $E_7(0,1,1)$ are significantly greater than 0. Therefore, only the following four ESS scenarios exist in this article, as shown in Table V.

Proposition 4: When $L < TP_1 + M_1 - R$, the pharmaceutical enterprise has a low expected return from building an UPMW. In this case, the pharmaceutical enterprise will choose not to build an UPMW, regardless of the consumer's and supplier's decisions, and the system reaches the equilibrium point $E_1(0,0,0)$. When $TP_1 + M_1 - R < L < M_1 + R + TP_1$, the expected revenue of pharmaceutical enterprises gradually increases, and the equilibrium state of the system will shift from $E_1(0,0,0)$ to $E_2(1,0,0)$. When $L > M_1 + R + TP_1,\ V_2 - Q_2 < V_1 - Q_1$,

TABLE V
FOUR SCENARIOS IN WHICH THE LOCAL EQUILIBRIUM POINTS BECOME ESS

	Equalizsation point	Conditions	Result
Scenario 1	$E_1(0,0,0)$	$-1/(M_1-L-R+TP_1)<0$ $-1/B<0$ $-1/C_3<0$	ESS
Scenario 2	$E_2(1,0,0)$	$1/(M_1-L+R+TP_1)<0$ $1/(Q_1-Q_2-V_1+V_2)<0$ $-1/C_3<0$	ESS
Scenario 3	$E_5(1,1,0)$	$1/(M_1-L+R+TP_1-KTP_2)<0$ $-1/(Q_1-Q_2-V_1+V_2)<0$ $-1/(C_3-\Delta P_2)<0$	ESS
Scenario 4	$E_8(1,1,1)$	$-\frac{1}{L-M_1-R-TP_1+TKP_2+\Delta P_1}<0$ $-1/(Q_1-Q_2-V_1+V_2+\rho V_2)<0$ $1/(C_3-\Delta P_2)<0$	ESS

the pharmaceutical enterprise's expected return from building an UPMW is higher, but the system reaches the equilibrium point $E_2(1,0,0)$due to the consumer's low benefit from using the UPMW. When $L>M_1+R+TP_1,\ V_2-Q_2>V_1-Q_1$, the utility of consumers using UPMW increases, and the equilibrium state of the system will evolve from $E_2(1,0,0)$ to other equilibrium states.

This situation is common in enterprises' strategic choice process. When $L<TP_1+M_1-R$, the managers of the enterprise generally perceive the UPMW as failing to generate sufficient benefits for the company, leading to a reluctance to invest. The stronger the organizational inertia, the lower the probability of innovation and the greater the willingness to maintain the status quo. When $L>M_1+R+TP_1$, the enterprise exhibits a robust propensity for innovation and a willingness to allocate resources to new technological investments. Notwithstanding the consumer's preference for abstaining from the UPMW, as evidenced by $V_2-Q_2<V_1-Q_1$, pharmaceutical enterprises in such scenarios prioritize future-oriented investment strategies and resolutely opt to establish an UPMW. To effectively elucidate the dynamic evolution trajectories of the aforementioned two equilibrium points, we specified the corresponding parameters and used MATLAB 2023b to perform the numerical simulation. The dynamic evolution of the equilibrium points $E_1(0,0,0)$ and $E_2(1,0,0)$ is illustrated in Figs. 2 and 3. The evolution probabilities of the tripartite converge to their respective equilibrium points, irrespective of the initial combination of states among the three parties.

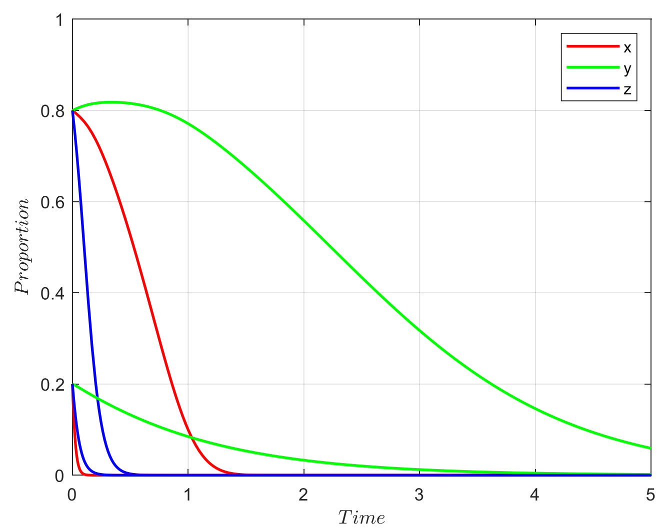


Fig. 2. Dynamic evolution path diagram in scenario 1. Note: $P_1=$ 1,000,000 yuan, $P_2=$ 1,400,000 $yuan$, $\Delta\ P_1=$ 50,000 yuan, $\Delta\ P_2=$ 100,000 Yuan, $V_1=4$, $V_2=4.5$, $Q_1=2$, $Q_2=2$, $C_1=$ 320,000 yuan, $C_2=$ 150,000 yuan, $C_3=$ 200,000 yuan, $L=$ 200,000 yuan, $B=1, T=0.5$, $K=1$, $M_1=300m000$ yuan, $R=0.5$, $\rho=0.2$.

Proposition 5: With the increase in consumers' willingness to use UPMW, it will also strengthen pharmaceutical enterprises' determination to construct UPMW, and the threshold value of the expected revenue for pharmaceutical enterprises to construct UPMW will change from M_1+R+TP_1 to $M_1+R+TP_1-KTP_2$.When $L>M_1+R+TP_1-KTP_2, V_2-Q_2>V_1-Q_1$, consumers will choose to use the UPMW as a more effective option than the traditional model. However, if $C_3>\Delta P_2$, the benefits related to the supplier's upgrading of equipment are lower than the costs; in this case, The system evolves from $E_2(1,0,0)$ to $E_5(1,1,0)$, with $E_5(1,1,0)$becomes the ESS point. When $C_3<\Delta P_2$, the supplier chooses to upgrade the equipment, and $L>M_1+R+TP_1-TKP_2-\Delta P_1,\ V_2-Q_2+\rho V_2>V_1-Q_1$; the pharmaceutical enterprise and consumer choose to build and use the UPMW, and thus, $E_8(1,1,1)$ is the ESS point.

Proposition 5 indicates that for the pharmaceutical enterprise, the threshold of expected benefits required to construct an UPMW decreases as consumers opt to use the UPMW and suppliers choose to upgrade equipment. Meanwhile, for customers,

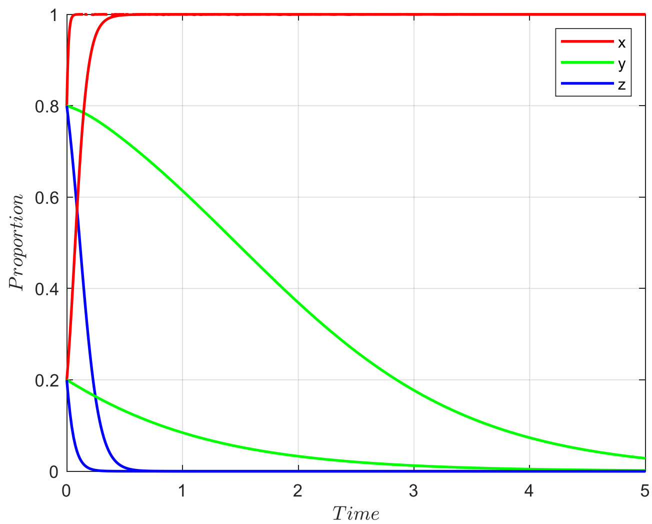


Fig. 3. Dynamic evolution path diagram in Scenario 2. Note: $P_1 = 1{,}000{,}000$ yuan, $P_2 = 2{,}000{,}000$ *yuan*, $\Delta\ P_1 = 50{,}000$ yuan, $\Delta\ P_2 = 100{,}000$ yuan, $V_1 = 4$, $V_2 = 3$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320{,}000$ yuan, $C_2 = 150{,}000$ yuan, $C_3 = 200{,}000$ yuan, $L = 800{,}000$ yuan, $B = 1$, $T = 0.5$, $K = 1$, $M_1 = 300{,}000$ yuan, $R = 0.5$, $\rho = 0.2$.

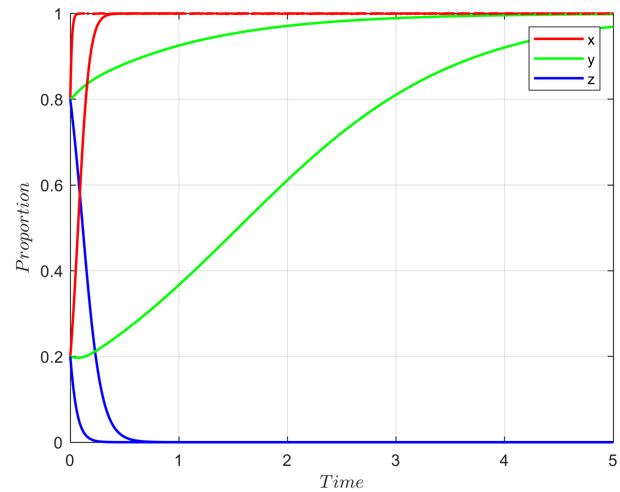


Fig. 4. Dynamic evolution path diagram in Scenario 3. Note: $P_1 = 1{,}000{,}000$ yuan, $P_2 = 2{,}000{,}000$ *yuan*, $\Delta\ P_1 = 50{,}000$ yuan, $\Delta\ P_2 = 100{,}000$ yuan, $V_1 = 4$, $V_2 = 5$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320{,}000$ yuan, $C_2 = 150{,}000$ yuan, $C_3 = 200{,}000$ yuan, $L = 800{,}000$ yuan, $B = 1$, $T = 0.5$, $K = 1$, $M_1 = 300{,}000$ yuan, $R = 0.5$, $\rho = 0.2$.

the threshold of initial utility required to adopt the UPMW decreases as suppliers choose to upgrade their equipment. This phenomenon can be explained using the framework of strategic complementarity and co-evolution as pharmaceutical enterprise managers, consumers and suppliers exhibit strong interdependence. The marginal benefit of any single participant is strictly constrained by the strategic choices of the other participants. Therefore, when consumers and suppliers align their decisions with those of pharmaceutical enterprises, the tripartite game generates a positive feedback effect. As the game operates as a dynamic process, the participants adjust their expected returns and modify their corresponding decisions upon observing the others' behaviors. To effectively demonstrate the participants' dynamic game process, we assign values to the corresponding parameters to satisfy the respective conditions. The evolution probability of the three parties converges to the corresponding equilibrium points, regardless of the parties' initial choices. The results are presented in Figs. 4 and 5.

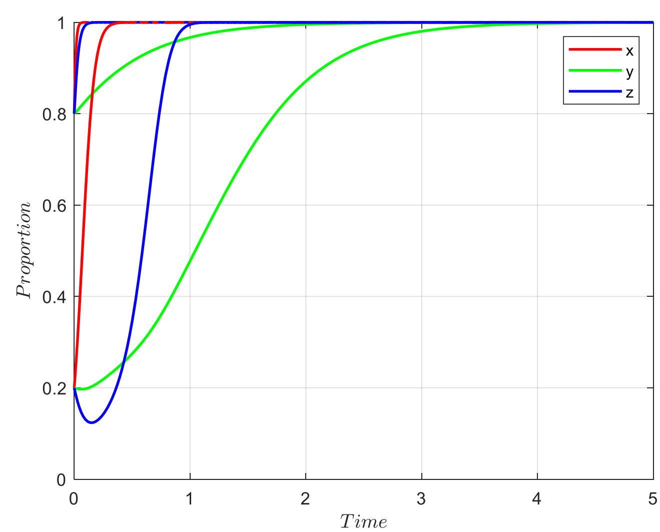
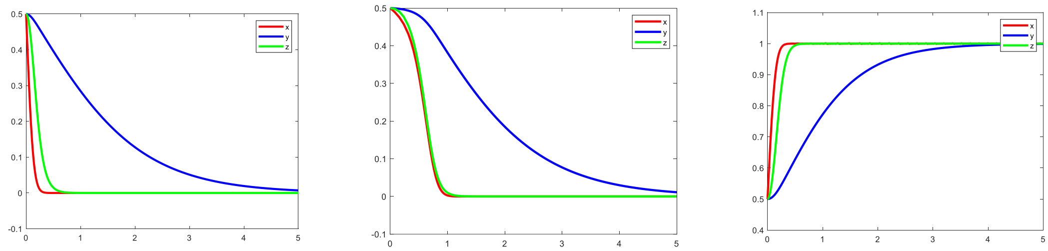


Fig. 5. Dynamic evolution path diagram in Scenario 4. Note: $P_1 = 1{,}000{,}000$ yuan, $P_2 = 2{,}000{,}000$ *yuan*, $\Delta\ P_1 = 100{,}000$ yuan, $\Delta\ P_2 = 600{,}000$ yuan, $V_1 = 4$, $V_2 = 5$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320{,}000$ yuan, $C_2 = 150{,}000$ yuan, $C_3 = 200{,}000$ yuan, $L = 800{,}000$ yuan, $B = 1$, $T = 0.5$, $K = 1$, $M_1 = 300{,}000$ yuan, $R = 0.5$, $\rho = 0.2$.

V. Simulation Analysis

In this section, we describe a numerical simulation conducted with reference to previous scholars [27], [53], [54]. Fahrenheit Pharmacy's UPMW is used as an example for testing the equilibrium solutions under different settings. Changes in the values of x, y and z can reflect the trend of system stability, and the participants' strategic tendencies can be represented. Stable values of x, y and z at a certain equilibrium point indicate that the game has reached system stability. When only one parameter is changed, such as precision service capability ρ, its impact on the game system can be revealed through the updated trajectories of x, y and z. In the numerical simulation, the evolutionary trends of x, y, and z also elucidate the strategic choice process wherein participants predict and react to each other's strategies [55].

In 2023, Fahrenheit Pharmacy officially landed in Shanghai, providing consumers with a new drug purchasing experience by disrupting the predicament of traditional pharmacies' heavy dependence on manpower and greatly improving operational efficiency. With reference to the case of Fahrenheit Pharmacy's UPMW and interview data collected by the research team, we set the following initial values for the numerical simulation: $P_1 = 100{,}000$ yuan, $P_2 = 2{,}000{,}000$ yuan, $\Delta\ P_1 = 100{,}000$ yuan , $\Delta\ P_2 = 400{,}000$ yuan, $V_1 = 4$, $V_2 = 4.5$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320{,}000$ yuan, $C_2 = 150{,}000$ yuan, $C_3 = 100{,}000$ yuan, $B = 1$, $M_1 =$



(a) (b) (c)

Fig. 6. Game evolution results from the three-party game system. (a) $L = 0$. (b) $L = 150\ 000$ yuan. (c) $L = 300\ 000$ yuan.

$300,000$ yuan and $R = 0.5$. The sensitivity analysis was carried out using MATLAB 2023b software, and the influences of different parameters on evolution were explored.

A. Influence of the Enterprise's Strategic Choice on the Decisions of the Three Parties in the Game

To further explore the influence of related variables on the game, a numerical evolution analysis was conducted based on the replication dynamic equation and the constraints of the optimal equilibrium point, focusing on the enterprise's and consumer's choices regarding the UPMW. MATLAB 2023b was used to conduct the evolutionary analysis, and the initial values were set as follows: $P_1 = 1,000,000$ yuan, $P_2 = 2,500,000$ yuan, $\Delta\ P_1 = 100,000$ yuan , $\Delta\ P_2 = 400,000$ yuan, $V_1 = 4$, $V_2 = 4.5$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320,000$ yuan, $C_2 = 150,000$ yuan, $C_3 = 100,000$ yuan, $B = 1$, $T = 0.5$, $K = 1$, $M_1 = 300,000$ yuan, $R = 0.5$ and $\rho = 0.2$. The initial decisions of *x*, *y* and *z* were 0.5, 0.5 and 0.5, respectively, in an exploration of how the future investment decisions of a pharmaceutical enterprise would affect the strategic choices of the enterprise and its consumers and suppliers. If pharmaceutical enterprises prioritize the opportunity cost of future development, then the opportunity cost L will be larger. Therefore, we assigned different values to the pharmaceutical enterprise's opportunity cost L for the UPMW, aiming to explore how pharmaceutical enterprises' development strategies influence on the game system.

As shown in Fig. 6(a), the opportunity cost for a pharmaceutical enterprise that does not consider its future strategy is 0. Given the unpredictability of consumer behavior, pharmaceutical enterprises maintain traditional models to maximize profits in the early strategic choice stage. Therefore, pharmaceutical enterprises are the first to decide to directly abandon the construction of an UPMW. Fig. 6(b) shows that pharmaceutical enterprises prioritize future strategies. The decision rates and decision paths of the pharmaceutical enterprise and supplier are shown to be similar in Fig. 6(a) and (b), indicating that pharmaceutical enterprises are cautious about the UPMW strategy and have a weak intention to build UPMWs. Suppliers, who are influenced by the strategic choices of pharmaceutical enterprises, tend to exhibit prudence in facility and equipment upgrading. Therefore, both sides emphasize short-term gains, which influences the evolution of the entire game system toward the state of (0,0,0). Fig. 6(c) shows that pharmaceutical enterprises prioritize the UPMW strategy and firmly choose to build UPMWs and focus on long-term benefits. Suppliers who align with the pharmaceutical enterprise's decision will also focus on future development and strive to improve their service capabilities, directing the evolution of the entire game system toward the state of (1,1,1).

B. Influence of the Proportion T *of an Enterprise's Goods Sold in the UPMW on the Game*

To further explore the influence of T on the evolutionary game, the following parameters were set: $P_1 = 1,000,000$ yuan, $P_2 = 2,500,000$ yuan, $\Delta\ P_1 = 100,000$ yuan , $\Delta\ P_2 = 400,000$ yuan, $V_1 = 4$, $V_2 = 4.5$, $Q_1 = 2$, $Q_2 = 2$, $C_1 = 320,000$ yuan, $C_2 = 150,000$ yuan, $C_3 = 100,000$ yuan, $B = 1$, $K = 1$, $L = 100,000$ yuan, $M_1 = 300,000$ yuan, $R = 0.5$, $\rho = 0.2$. By substituting $T = 0.2,\ 0.5,\ 0.8$ into the formula, the influence of the size of T on system evolution could be determined, as shown in Fig. 7.

As shown in Fig. 7, as T increases, the strategic choice window of the three parties decreases. Therefore, increasing the proportion of UPMW sales helps to shorten the acceptance cycle, making consumers more inclined to use UPMWs. For suppliers, upgrading their service capabilities also leads to higher returns; therefore, suppliers also choose to upgrade UPMWs. Further analysis shows that when the scale and service capabilities of an enterprise's UPMW are significantly improved, consumers are increasingly likely to choose the UPMW's services from among the available shopping methods. This occurs because improved service capabilities often result in a more convenient, efficient and secure shopping experience that effectively meets consumers' diverse needs and encourages them to use the UPMW's services. However, if enterprises provide low UPMW service coverage (e.g., limited service scope, poor service quality, other inconveniences), consumers are more inclined to choose traditional shopping methods. This occurs because such consumers may not gain sufficient convenience or value from UPMW services and thus prefer to retain their original shopping habits.

C. Influence of UPMW Consumer Acceptance K *on the Game Process*

To further explore the influence of K on the evolutionary game process, the following parameters were set: $P_1 = 1,000,000$ yuan, $P_2 = 2,500,000$ yuan, $\Delta\ P_1 =$

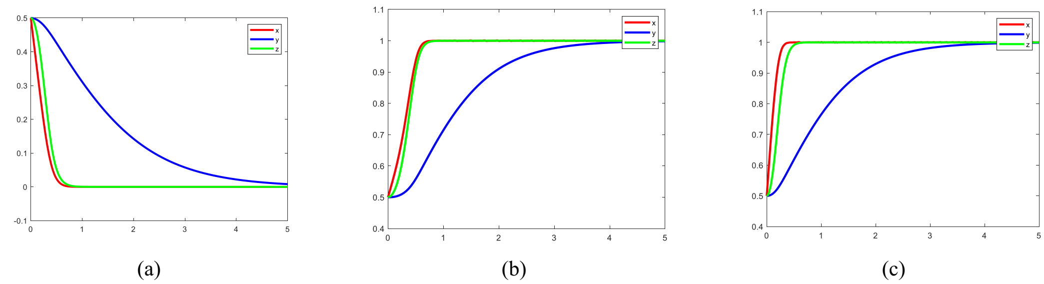


Fig. 7. Effect of an UPMW on the sales proportion T in the evolutionary game system. (a) $T = 0.2$. (b) $T = 0.5$. (c) $T = 0.8$.

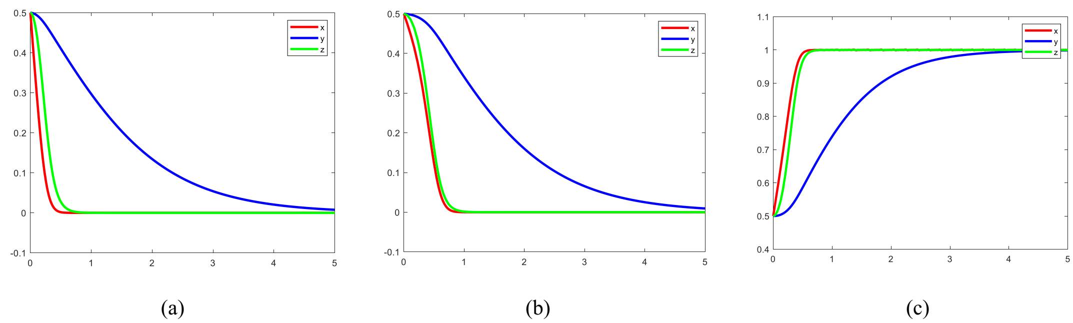


Fig. 8. Influence of consumer acceptance K on the evolutionary game system. (a) $K = 0.8$. (b) $K = 0.9$. (c) $K = 1$.

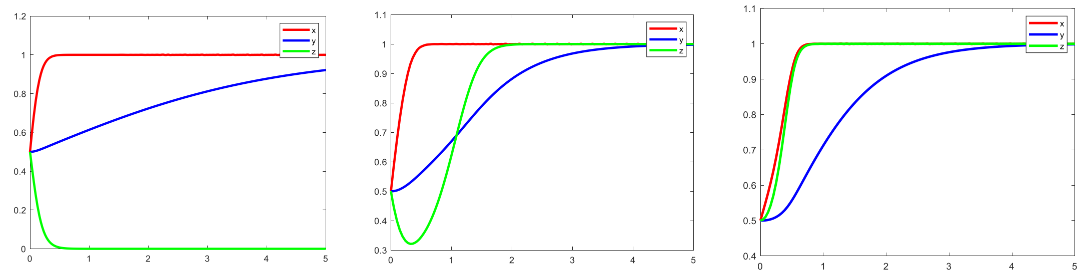
$100,000$ yuan , $\Delta \ P_2 = \ 400,000$ yuan, $V_1 = \ 4$, $V_2 = 4.5$, $Q_1 = \ 2$, $Q_2 = \ 2$, $C_1 = \ 320,000$ yuan, $C_2 = 150,000$ yuan, $C_3 = \ 100,000$ yuan, $B = 1$, $T = 0.5$, $L = 200,000$ yuan, $M_1 = \ 300,000$ yuan, $R = 0.5$, $\rho = 0.2$. $K = 0.8, \ 0.9, \ 1$ could then be substituted into the formula to determine the influence of K on system evolution, as shown in Fig. 8.

As shown in Fig. 8(a)–(c), as K decreases, the willingness of the pharmaceutical enterprise to build an UPMW also decreases significantly. As shown in Fig. 8(a), when consumer acceptance of UPMWs is low, the market for UPMWs is small. In this case, the pharmaceutical enterprise chooses not to build an UPMW and instead maintain their traditional pharmacies to obtain maximum profits. Fig. 8(b) demonstrates that as consumer acceptance increases, the duration of the pharmaceutical enterprise's delay in adopting UPMWs also increases, indicating that excessively low consumer acceptance leads pharmaceutical enterprises to abandon UPMW construction. Fig. 8(c) shows that when consumer acceptance of UPMWs is high, the consumer market stabilizes. The pharmaceutical enterprise's confidence in UPMW construction increases, and the enterprise firmly chooses to build an UPMW; here, the supplier's state also evolves to (1,1,1). This indicates that when the consumer market is less receptive to new technologies, enterprises tend to retain their original business model, which reflects the risk aversion theory of enterprise management. When adopting new technologies, pharmaceutical enterprises face increases in various risks, including technical risk, market risk and consumer acceptance risk. To mitigate these risks, pharmaceutical enterprises may choose to maintain their original business model to ensure the stability and sustainability of their business.

D. Influence of Pharmaceutical enterprises' Subsidies to Suppliers on the Game Process

As UPMWs represent an emerging industry and new technology, their successful implementation can enhance an enterprise's stock price and social image, providing additional benefits for both the enterprise and its suppliers. However, suppliers' willingness to upgrade may be low, given the potential for a delay in generating additional revenue during the early stage of UPMW construction. To promote UPMW construction and development, pharmaceutical enterprises aim to increase suppliers' willingness to upgrade by offering financial support. To explore the effect of such support, the values of P_2, ΔP_1, andΔP_2 can be reassigned and the other parameters set as follows: $P_1 = 1\,000\,000$ yuan, $\Delta P_1 = 100\,000$ yuan , $V_1 = 4$, $V_2 = 4.5$, $Q_1 = \ 2$, $Q_2 = \ 2$, $C_1 = \ 320\ 000$ yuan, $C_2 = 150\ 000$ yuan, $C_3 = \ 100\ 000$ yuan, $B = 1$, $T = 0.5$, $L = 100\ 000$ yuan, $M_1 = \ 300\ 000$ yuan, $R = 0.5$ and $\rho = 0.2$. The simulation results are shown in Fig. 9.

As shown in Fig. 9(a)–(c), the subsidies provided to the suppliers of pharmaceutical enterprises help to promote the evolution of the game system to (1,1,1). These subsidies increase the willingness of the suppliers to upgrade, thus helping to improve the UPMW's service capabilities. In Fig. 9(a), the pharmaceutical enterprise does not subsidize the supplier. In this scenario, the supplier chooses not to upgrade the equipment because the associated benefits would not compensate for the cost of the equipment upgrade. In Fig. 9(b), the pharmaceutical enterprise partially compensates the supplier. Given the small size of the subsidy, however, the supplier remains unlikely to upgrade the facility and equipment in the early stage. However, when the pharmaceutical enterprise firmly chooses to assume UPMW,



(a) (b) (c)

Fig. 9. Impact of pharmaceutical enterprises' subsidies to suppliers on the gaming system. (a) $P_2 =$ 2 500 000 yuan, $\Delta P_1 =$ 10 000 yuan, $\Delta P_2 = 0$. (b) $P_2 =$ 2 300 000 yuan, $\Delta P_1 =$ 10 000 yuan, $\Delta P_2 =$ 20 000 yuan. (c) $P_2 =$ 2 100 000 yuan, $\Delta P_1 =$ 10 000 yuan, $\Delta P_2 =$ 40 000 yuan.

the supplier also chooses to upgrade the facility and equipment. Fig. 9(c) shows that suppliers that receive higher subsidies are more likely to follow the pharmaceutical enterprise's decision, and the evolution speed of both the pharmaceutical enterprises and suppliers is accelerated compared with that in Fig. 9(b). A comparison of Fig. 9(a)–(c) reveals that an increase in subsidies provided by pharmaceutical enterprises also helps to increase consumers' willingness to choose UPMWs.

VI. Conclusion and Future Studies

A. Research Conclusions

This article refines and establishes an evolutionary game framework of UPMW adoption behavior that involves enterprises, consumers, and suppliers. This framework reveals the evolutionary process of the tripartite decision-making and clarifies the impact of such decision-making on the game system. By applying equilibrium theory, the article conducts an in-depth analysis of the locally stable equilibrium results in the tripartite game process, uncovering the intrinsic logic behind the dynamic changes of the game. Furthermore, key variables are selected for numerical simulation analysis and scenario expansion analysis of cooperative R&D between pharmaceutical enterprises and suppliers, which reveals the mechanism by which relevant parameters influence the game system during the tripartite game. This study has addressed the research questions proposed herein and drawn the following core conclusions.

First, during UPMW construction, managers of pharmaceutical enterprises exhibit a nonprofit-oriented nature, the preference of pharmaceutical enterprise managers plays a decisive role, rather than being determined solely by whether the project is profitable. When managers implement aggressive strategies, pharmaceutical enterprises resolutely invest in UPMW construction, which can easily lead to a situation where UPMWs fail to meet consumer demand and pharmaceutical companies suffer losses. However, when managers adopt conservative strategies, their decision is not affected by the decisions of consumers and suppliers, and the enterprise resolutely refrains from constructing UPMWs. However, this choice also may result in a loss of competitive advantages for pharmaceutical enterprises. When a pharmaceutical enterprise adopts an ambiguous strategic stance, its strategic choice is influenced by the preferences of its consumers and suppliers. Specifically, consumers' and suppliers' positive strategic choices can incentivize a pharmaceutical enterprise to construct an UPMW.

Second, consumers use an UPMW only when the utility difference between the UPMW and the traditional model exceeds the difference between the attention cost and potential additional utility, merely improving the user experience of UPMW may not be able to attract consumers to use UPMW. Suppliers' decisions are based on the difference between the costs and benefits of upgrading, and their acceptable cost threshold declines as the pharmaceutical enterprise's construction willingness and consumers' usage intention increase. Through numerical simulations, we show that improving the consumer experience and subsidizing suppliers can accelerate the evolution of the game system toward an optimal state. When subsidies are limited, suppliers only opt to upgrade their equipment if the pharmaceutical enterprise resolutely commits to constructing UPMWs (i.e., when the decision probability $x = 1$). Otherwise, suppliers refrain from upgrading. When the enterprise decides to build an UPMW, consumers decision regarding its use is driven by demand, which exemplifies the "follow-up effect" in market supply–demand dynamics. Meanwhile, suppliers' decision to upgrade UPMW facilities is driven by additional revenue.

Third, the precise service capability, sales proportion and acceptance level of the UPMW are key variables in the game system. If these values are significantly low, they shift the system from constructing to not constructing an UPMW. In addition, the UPMW sales value is directly related to the system evolution rate. Either an increase or decrease in this value accelerates the evolution of the entire evolutionary game system toward one of two extreme states: complete rejection or total adoption by all three parties, reflecting the dual impact of market acceptance and technology maturity on the promotion of emerging business formats. Furthermore, low customer acceptance directly leads the pharmaceutical enterprise to rely on a traditional business model. Consumers are affected by herd mentality; therefore, when UPMW market acceptance is low, its benefits decline, forcing the pharmaceutical enterprise to abandon the UPMW in favor of a traditional business model.

In summary, this study deepens our understanding of the interaction mechanisms between enterprises, consumers and suppliers with respect to UPMW adoption behavior. It also provides a scientific basis for enterprise managers seeking to effectively promote the development of the UPMW market through quantitative analysis.

B. Implications

1) Theoretical Implications: Following an in-depth analysis of the evolutionary game model representing the UPMW adoption behavior of pharmaceutical enterprises, suppliers and consumers, we draw the following theoretical contributions.

First, prior research on unmanned retail focuses on the retail industry (e.g., unmanned stores, vending cabinets), while research on pharmaceutical retail mostly concerns traditional issues such as pricing and drug quality. Few studies have explored the feasibility of the emerging business model of UPMWs. The current study is the first to incorporate UPMWs into a theoretical analysis framework for pharmaceutical retail. By exploring the technical adaptability and cost effectiveness of UPMWs and industry demands in pharmaceutical distribution scenarios, this article provides a new theoretical perspective on prompting innovation in pharmaceutical retail.

Second, traditional research on unmanned retail has focused on the factors influencing consumers' experiences and usage after implementation but has lacked theoretical explanations of enterprises' strategic choices during construction. In contrast, our study systematically explores influential factors such as technological maturity and market demand on decisions regarding UPMW construction from the perspective of pharmaceutical enterprises, thus filling a gap in research on unmanned retail at the construction stage.

Third, the literature pays little attention to the dynamic game relationship involving enterprises, consumers, and suppliers in unmanned retail development. In the current study, we constructed a tripartite evolutionary game model comprising pharmaceutical enterprises, consumers and suppliers. By analyzing these players' strategic choices and evolutionary paths during the promotion of UPMWs, we revealed the mechanism of coordination among the parties during the diffusion of emerging business forms. Furthermore, our study introduces EGT into research on UPMWs. By dynamically simulating the iterative process of the main body's strategy, our study reflects the complexity of the decisions made by all parties in reality.

2) Managerial Implications for Managers of Pharmaceutical Enterprises: In addition to the theoretical contributions, this article contains an in-depth analysis of our evolutionary game model of the UPMW adoption behaviors of both pharmaceutical enterprises and suppliers. For pharmaceutical enterprises, the following measures can be adopted.

First, it is necessary to establish a diversified decision-making reference system and introduce a demand feasibility assessment link. By using tools such as market research and consumer demand forecasting models, enterprises can predict whether UPMW can match market demand after its construction, so as to avoid losses caused by blind construction. Consumer willingness and supplier willingness should be incorporated into core decision-making indicators. Enterprises can establish a consumer demand feedback platform (e.g., online questionnaires, offline experience surveys) and a supplier communication mechanism (e.g., regular seminars, cooperation intention collection), convert their positive feedback into specific data indicators, and form a "Construction Feasibility Score" through weighted calculation. Based on this score, enterprises can formulate UPMW construction decisions, thereby avoiding wavering decisions caused by ambiguous attitudes.

Second, targeting the decision-making logic of consumers in the UPMW development phase, enterprises should optimize services from the dimension of utility difference. They can widen the utility gap between UPMW and traditional models through functional upgrades—for instance, increasing the coverage of drug types, optimizing the drug pickup process, and providing additional services. Meanwhile, enterprises can leverage the "bandwagon effect" to expand the user base and activate the market through word-of-mouth communication from early users. Specifically, they can select seed users (e.g., young groups with high acceptance of new things, community residents with emergency drug purchase needs), provide free trials, exclusive benefits, and other incentives to encourage them to share their usage experiences. In addition, in UPMW coverage areas, enterprises can conduct scenario-based promotion (e.g., via community bulletin boards, online local lifestyle platforms), display usage cases of other consumers, and utilize conformity psychology to attract more users to try UPMW.

Third, enterprises should establish a UPMW operation data monitoring system to track indicators such as precision service capabilities and sales proportion in real time, and rate consumers' usage experiences. When consumers perceive low precision service capabilities, enterprises should invest more in technological R&D resources—for example, optimizing the algorithm for AI-based intelligent drug recommendations, improving inventory management systems to enhance response speed, and boosting drug delivery accuracy and timeliness. When the sales proportion is at a low level, enterprises should timely adjust operational strategies, such as optimizing UPMW locations and adding new UPMW facilities.

3) Managerial Implications for Managers of Suppliers: As manufacturers of UPMW facilities and supply chain partners of pharmaceutical enterprises, the following actions can be taken by suppliers.

First, suppliers should prioritize long-term strategic development. As a new format in pharmaceutical retail, an UPMW must undergo long-term improvements. Therefore, suppliers should focus on the future benefits and enhance the UPMW's service capabilities through continuous improvement measures. At the operational level, suppliers should increase their R&D investment and develop equipment and technologies tailored to scenarios such as online–offline integrated sales and personalized health management in accordance with the business format requirements of an UPMW. Suppliers should also fully participate in the construction of the UPMW ecosystem and collaborative with pharmaceutical enterprises to explore innovative marketing models, such as joint live broadcasts of health science and customized health service packages, to enhance user stickiness. From the perspective of long-term benefits, suppliers who take the lead in adapting to the UPMW model not only can seize market opportunities but also are able to form deep strategic bonds with pharmaceutical enterprises through continuous brand building and service upgrades. These suppliers can secure a more stable customer base, improve their brand premium and voice in the industry and realize the leap from short-term transactions to value symbiosis.

Second, suppliers should leverage consumers' acceptance to enhance the competitiveness of their own equipment. They can conduct market research to identify consumers' pain points when using UPMW equipment (e.g., complicated operations, insufficient functions) and optimize equipment design in a targeted manner—such as adding voice navigation functions and simplifying operation steps. Meanwhile, suppliers can convey the advantages of upgraded equipment (e.g., faster drug pickup speed, more accurate drug management) to consumers through pharmaceutical enterprises, so as to improve consumers' recognition of their own equipment. This will thereby enhance the cooperation stickiness with pharmaceutical enterprises and help suppliers gain an advantageous position in market competition.

In addition to the enterprises' efforts, governments and relevant institutions should provide additional policy guidance and support to create a favorable external environment for UPMW development. Specifically, governments could introduce policies that encourage and support enterprises in building UPMWs, such as tax incentives and financial support.

C. Future Research

Future studies can further explore the construction of UPMWs. First, such research should consider the government as the overseer of emerging businesses. A four-party evolutionary game model that includes the government, pharmaceutical enterprises, suppliers and consumers could be developed to further explore the construction of UPMWs. Second, this article does not fully address the action rules between pharmaceutical enterprises and suppliers nor the contract mechanism between enterprises. Future research could consider the effects of various contract types, such as incentive contracts and revenue sharing contracts, on the outcome of the game.

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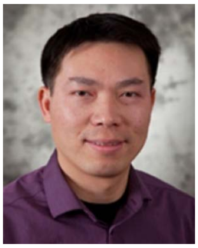

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